

Yaohui Shu

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RESEARCH OVERVIEW

My research has two connected threads: stochastic modeling of the physical processes that set the cloud droplet size distribution (CDS) in turbulence, and causal inference for aerosol-cloud-turbulence interactions. Supersaturation in turbulent clouds fluctuates over a wide range of time scales, so droplets with different supersaturation histories grow at different rates and the CDS broadens beyond the prediction of mean-field parcel theory. I represent condensational growth, the fluctuating saturation ratio, and entrainment-mixing as coupled stochastic processes in Langevin and Fokker-Planck form, with liquid water content and droplet number concentration conserved, and study how the correlation time and intensity of supersaturation fluctuations control spectral broadening and relative dispersion, through analytical limits and large Lagrangian Monte Carlo ensembles.

The second thread treats those same models as testbeds for causal inference. Aerosol-cloud-turbulence interactions form a high-dimensional, nonlinear system: aerosol, droplet, thermodynamic, and turbulence variables are coupled over a wide range of time and space scales, the couplings run in both directions, and the drivers that matter are often unmeasured, so correlation-based diagnostics cannot say which process forces which. I embed information-theoretic measures of directed influence (Liang-Kleeman information flow, transfer entropy, conditional mutual information) in stochastic condensation models whose causal structure is known analytically, and use them as ground truth for causal discovery methods that tolerate feedback and unobserved drivers, before applying those methods to observations and simulations. The longer-term goal is a process-level, memory-aware understanding of droplet spectral evolution that can inform microphysics parameterizations.

EDUCATION

Ph.D., Applied Mathematics and Statistics 2022 – May 2027 (expected)

Stony Brook University, Stony Brook, NY

Dissertation: *Uncovering Causal Mechanisms of Aerosol-Cloud-Turbulence Interactions Using Physics- and Data-Driven Methods*

Advisors: Xiaolin Li (Stony Brook University) and Yangang Liu (Brookhaven National Laboratory)

Advanced to candidacy, Fall 2025. GPA: 3.78 / 4.0

Coursework: Advanced Stochastic Models, Differential Equations I and II, Numerical Analysis I–III, Principles of Parallel Computing, Applications of Complex Analysis.

M.S., Finance, University of Rochester, Rochester, NY. GPA: 3.81 / 4.0 Aug 2021 – Aug 2022

Certificate in Quantitative Finance (CQF), CQF Institute

RESEARCH EXPERIENCE

Doctoral Researcher, Stony Brook University Aug 2023 – Present

Research Assistant, Brookhaven National Laboratory Jan 2024 – Jan 2026

Sole Ph.D. student on a joint program of the Department of Applied Mathematics and Statistics (Stony Brook) and the Environmental Science and Technologies Department (Brookhaven) on stochastic condensation and causal analysis of aerosol-cloud-turbulence interactions, responsible for research direction, theory, code, and manuscripts. Supported by the DOE Atmospheric System Research (ASR) Program and BNL Laboratory Directed Research and Development (LDRD-25-008).

Memory of supersaturation fluctuations and spectral broadening

- Generalized the McGraw and Liu (2006) Brownian drift-diffusion model, valid only in the strongly damped limit, by representing the saturation ratio as an Ornstein-Uhlenbeck process coupled to the

condensational growth equation. The decorrelation rate spans the full memory continuum, and the McGraw-Liu model is recovered as it goes to infinity [1].

- Lagrangian Monte Carlo simulations conserving liquid water content and droplet number concentration, with time-step and ensemble-size convergence tests, show that relative dispersion increases with both the correlation time and the standard deviation of supersaturation fluctuations, that the white-noise model underestimates broadening at long correlation times, and that the dependence collapses on the ratio of fluctuation intensity to decorrelation rate [1].

Information flow between supersaturation and droplet growth

- Incorporated the Liang-Kleeman information-flow formalism into the coupled model, derived the supersaturation-to-droplet information flow in closed form with its white-noise and frozen-noise limits, and estimated it from ensemble covariances across droplets. The estimated flow matches theory, the reverse flow vanishes, and the causal influence of supersaturation on droplet size weakens as the CDS broadens, a directional statement the symmetric correlation coefficient cannot make [1].

Maximum entropy systems theory and Langevin stochastic condensation

- Maximum entropy systems theory predicts a Weibull CDS whose order follows from the conserved quantity but has no dynamics; Langevin stochastic condensation has dynamics, but its spectral order is fixed by the variable in which the noise is additive. Showed that their disagreement under mass conservation reflects the choice of size variable, and generalized stochastic condensation to an arbitrary conserved moment order under the equal-shares hypothesis, imposed through the growth law, the saturation ratio a droplet experiences, or both [2].
- Showed that the conserved order sets the spectral shape while the normalizing moment sets only the scale; Monte Carlo simulations across conserved orders reproduce the predicted narrowing. The residual underestimate of relative dispersion traces to the small-size boundary, where in the finite-memory model no rule can conserve probability, leave the saturation ratio unchanged, and avoid accumulation at zero size. The three formulations share a size spectrum but predict different joint statistics of droplet size and local saturation ratio, so observations can discriminate among them [2].

Entrainment-mixing in a stochastic condensation framework

- Built a three-state stochastic condensation and entrainment-mixing model: an adiabatic steady state, an entrainment event by unbiased thinning of the droplet population, and extreme-inhomogeneous (complete evaporation) and homogeneous (uniform shrinkage) pathways to the same target liquid water content. The pathways produce distinct spectral evolutions and trajectories of the homogeneous mixing degree measures of Lu et al. (2013) [3].
- Formulated a general entrainment-mixing operator that spans the continuum between the two limits and generates controlled fluctuations of droplet number concentration and liquid water content, to probe how entrainment-induced microphysical variability reshapes the CDS and to constrain mixing parameterizations [3].

Causal discovery with feedback and unobserved drivers

- Developing a causal discovery framework for high-dimensional, nonlinear spatiotemporal geophysical data that assumes neither acyclicity nor causal sufficiency: partial ancestral graphs over a lag window under sigma-separation, nearest-neighbor conditional mutual information tests, orientation from time precedence and nonstationarity, a surrogate-calibrated diagnostic that separates nonlinearity from nonstationarity, and a path-space relative entropy rate (closed form by Girsanov's theorem) as a time-resolved measure of causal strength. Validation uses stochastic condensation systems in which the presence and absence of directed influence are known [4].

RESEARCH PLAN

- *Two-way coupling and vapor-liquid feedback.* Close the coupling between supersaturation and droplet growth so that condensation depletes supersaturation, which enables causal attribution of vapor-liquid feedback and provides analytically controlled feedback benchmarks for causal discovery.
- *Activation and deactivation.* Add curvature and solute effects to represent aerosol activation and droplet

deactivation at the small-size end, replacing idealized boundary closures, and quantify how this closure and the imposed conservation constraints shape the CDS.

- *Entrainment-mixing and supersaturation variability.* Couple the general entrainment-mixing operator to finite-memory supersaturation fluctuations, linking mixing regimes, fluctuations of droplet number and liquid water, and spectral broadening within one stochastic framework.
- *Nonlinear causal attribution.* Move beyond linear-Gaussian information-flow estimators by embedding transfer entropy and conditional mutual information in causal machine learning, for the nonlinear and non-Gaussian regimes typical of turbulent clouds.
- *Observational constraints.* Test memory-dependent broadening, information transfer, and the predicted joint statistics of droplet size and supersaturation against laboratory cloud chamber and field observations, including long-term ARM measurements, toward memory-aware microphysics parameterizations.

PUBLICATIONS

Journal articles

- [1] **Shu, Y.**, Liu, Y., Zhang, T., McGraw, R., and Li, X., 2026: Stochastic condensation with information flow for causality analysis. *Geophysical Research Letters*, **53**(17), e2026GL123056. doi:10.1029/2026GL123056. Open access.

Manuscripts in preparation

- [2] **Shu, Y.**, Liu, Y., and Li, X.: A Langevin-type stochastic condensation model to explain the size distribution based on maximum entropy principle. For *Physical Review E*.
- [3] **Shu, Y.**, Liu, Y., and Li, X.: A new stochastic condensation model for investigating turbulent entrainment-mixing processes. For *Journal of the Atmospheric Sciences*.
- [4] **Shu, Y.**: Entropy-based causal discovery for spatiotemporal dynamical systems with feedback and unmeasured drivers. Methodology working paper.

Software

- `stochastic_condensation_info_flow` (v1.2), code for [1]. MIT License. doi:10.5281/zenodo.22089240; GitHub.

CONFERENCE ABSTRACTS AND PRESENTATIONS

- Liu, Y., **Shu, Y.**, Zhang, T., and McGraw, R., 2026: A new generalized stochastic model and impacts of microphysical fluctuations. *106th AMS Annual Meeting*, January 2026 (presented by Y. Liu).
- **Shu, Y.**, Liu, Y., Zhang, T., and McGraw, R. L., 2025: Stochastic condensation/evaporation with different physical constraints and information flow for causal discovery. Poster, *AGU Fall Meeting 2025*, A21B-2034.
- Liu, Y., Gao, S., Lu, C., and **Shu, Y.**, 2025: Relationship between turbulent entrainment-mixing and droplet clustering. *AGU Fall Meeting 2025*, A21B-203.
- Liu, Y., **Shu, Y.**, Zhang, T., and McGraw, R., 2025: Stochastic condensation with information flow for causality analysis. Poster, *DOE Atmospheric System Research (ASR) Program Meeting*, 2025.

TEACHING

Teaching Assistant, AMS 595 Fundamentals of Computing, Stony Brook University

Fall 2026

TECHNICAL SKILLS

Computing: Python (NumPy, SciPy, pandas), MATLAB, C++, R, Git; SLURM clusters and parallel ensemble workflows.

Methods: Lagrangian Monte Carlo, SDE integrators, Itô calculus, Langevin and Fokker-Planck equations, maximum entropy, information flow, transfer entropy, conditional mutual information, causal discovery.